

Week 2: TA Problems and Solutions

Linear Algebra I · Fall 2026

15 alternative problems in increasing difficulty.

Suggested 80-minute session: work through core problems 1-12 and use problems 13-15 as extensions according to progress.

All vectors use column notation. The notebook retains executable Python for the figures; the PDF omits the code.

Practice problems

The theory section ends here.

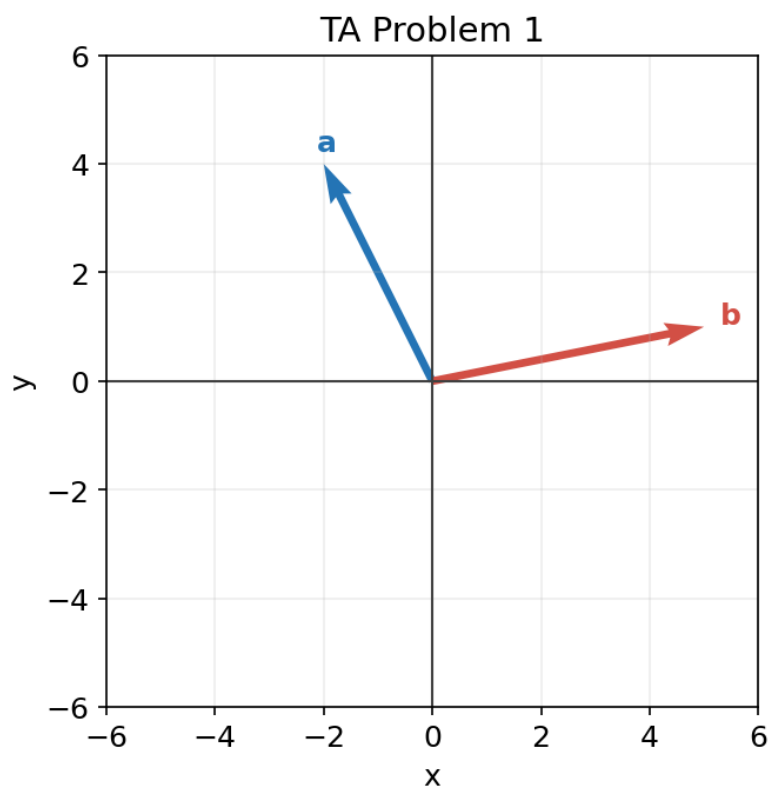
Try each problem before reading its solution. A wide blank space separates the statement from the answer. Each problem starts a new page in the PDF.

Problem 1: Dot product

$\mathbf{a} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$. Compute $\mathbf{a} \cdot \mathbf{b}$ and classify the angle.

Solution 1

$\mathbf{a} \cdot \mathbf{b} = -6 < 0$, so the angle is obtuse.



Problem 2: Norm and unit vector

Find the norm and a unit vector in the direction of $\mathbf{v} = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$.

Solution 2

$$\|\mathbf{v}\| = 3, \text{ so } \hat{\mathbf{v}} = \begin{bmatrix} -2/3 \\ 1/3 \\ 2/3 \end{bmatrix}.$$

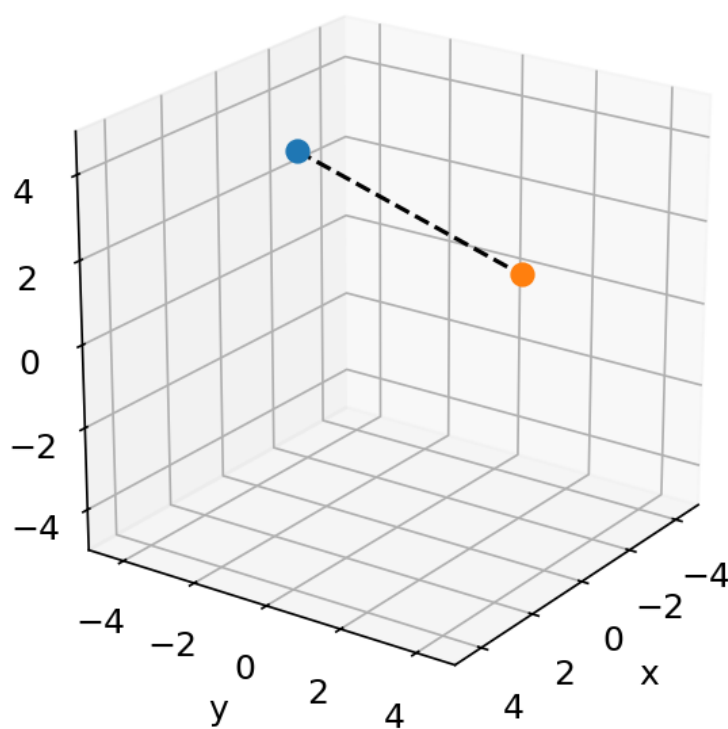
Problem 3: Distance

Find the distance between $\mathbf{p} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$ and $\mathbf{q} = \begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$.

Solution 3

$\mathbf{p} - \mathbf{q} = \begin{bmatrix} 3 \\ -4 \\ 3 \end{bmatrix}$, so the distance is $\sqrt{34}$.

Distance between p and q



Problem 4: Unknown for orthogonality

Find t if $\begin{bmatrix} 1 \\ t \\ 2 \end{bmatrix}$ is orthogonal to $\begin{bmatrix} 2 \\ -3 \\ 4 \end{bmatrix}$.

Solution 4

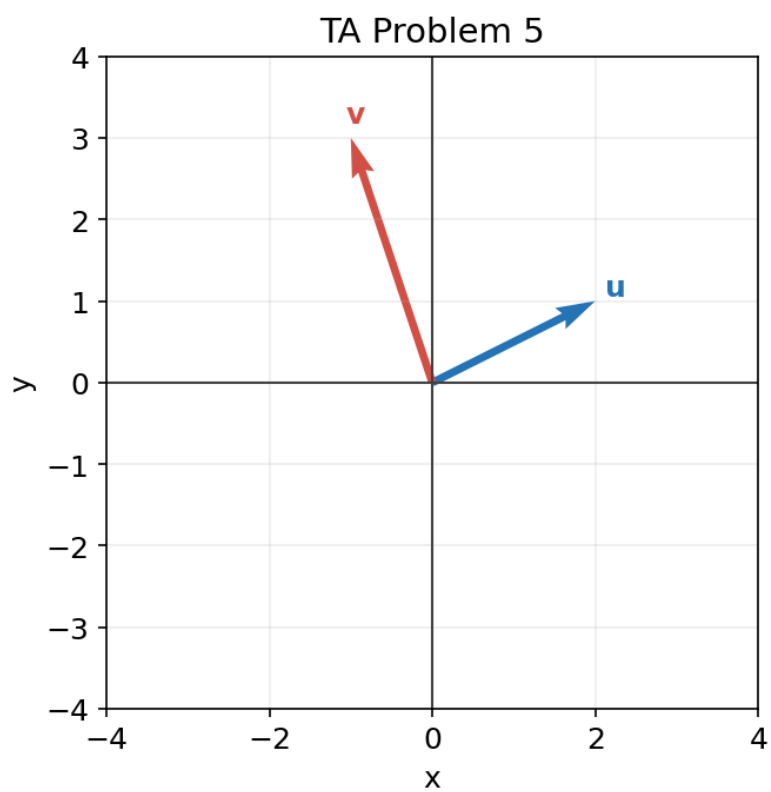
$2 - 3t + 8 = 0$, so $t = 10/3$.

Problem 5: Angle

Find $\cos \theta$ for $\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$.

Solution 5

$$\cos \theta = 1/\sqrt{50} = \sqrt{2}/10.$$

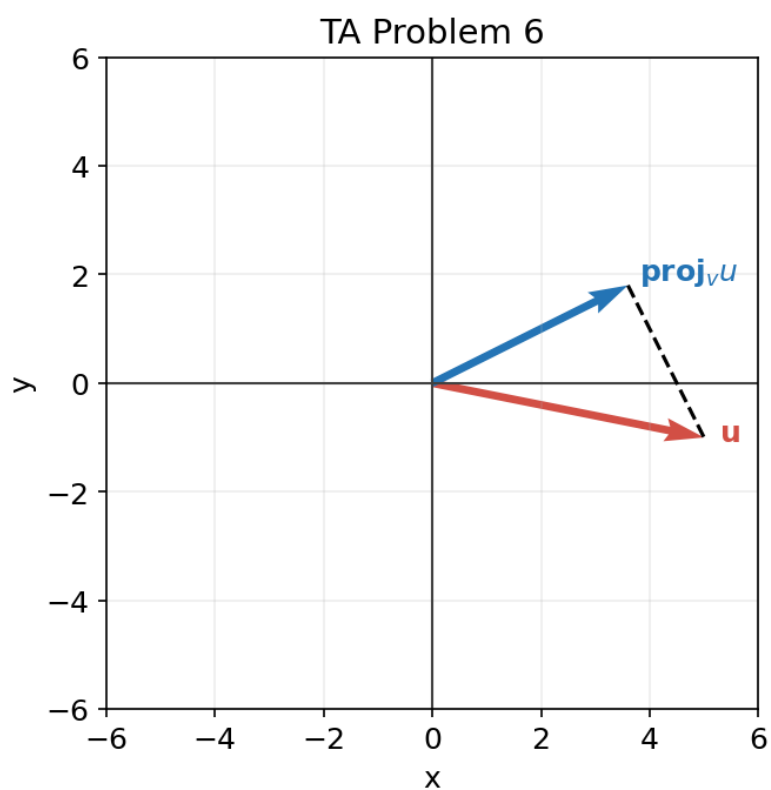


Problem 6: Projection

Project $\mathbf{u} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$ onto $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Solution 6

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \begin{bmatrix} 18/5 \\ 9/5 \end{bmatrix}.$$



Problem 7: Orthogonal decomposition

Write $\mathbf{u} = \begin{bmatrix} 5 \\ -1 \end{bmatrix}$ as a part parallel to $\mathbf{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ plus a part orthogonal to $\mathbf{v}$.

Solution 7

$$\mathbf{u} = \begin{bmatrix} 18/5 \\ 9/5 \end{bmatrix} + \begin{bmatrix} 7/5 \\ -14/5 \end{bmatrix}.$$

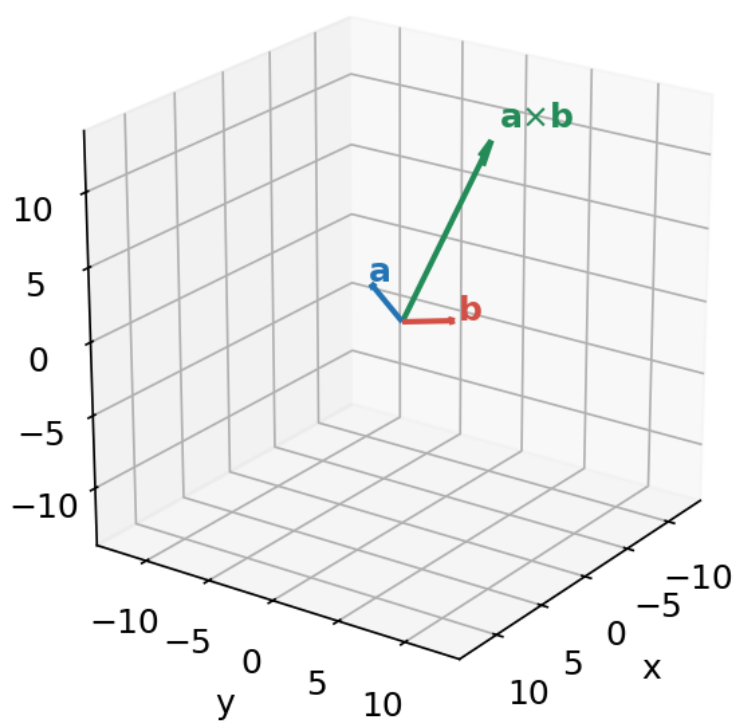
Problem 8: Cross product

Compute $\mathbf{a} \times \mathbf{b}$ for $\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 0 \\ 4 \\ 1 \end{bmatrix}$.

Solution 8

$$\mathbf{a} \times \mathbf{b} = \begin{bmatrix} -13 \\ -2 \\ 8 \end{bmatrix}.$$

TA Problem 8



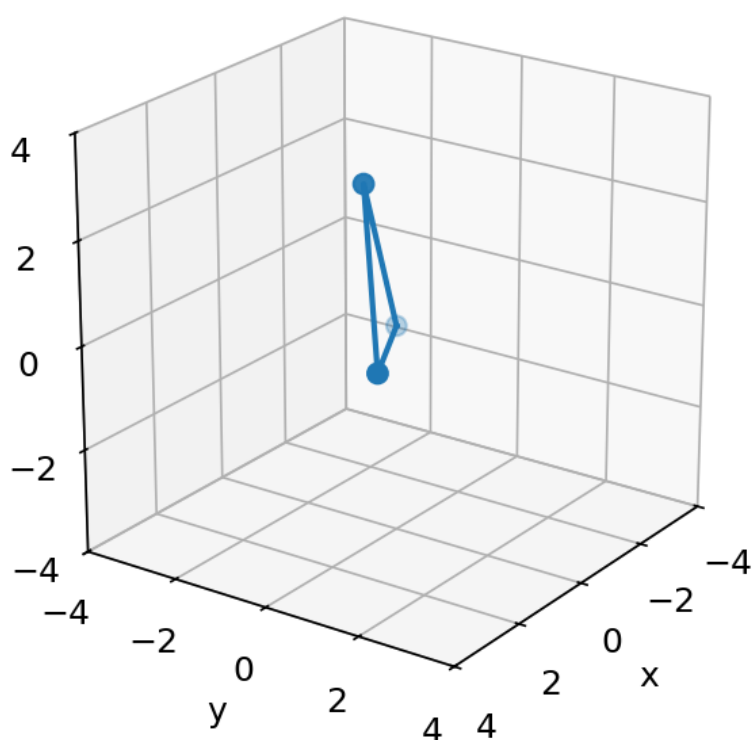
Problem 9: Triangle area

$A = (0, 0, 0)$, $B = (2, 1, 0)$, $C = (1, 0, 3)$. Find the triangle area.

Solution 9

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{bmatrix} 3 \\ -6 \\ -1 \end{bmatrix}, \text{ so the area is } \sqrt{46}/2.$$

Triangle ABC



Problem 10: Matrix basics

For $M = \begin{bmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{bmatrix}$, state its size, m_{21} , and all columns.

Solution 10

M is 2×3 , $m_{21} = 4$, and the columns are $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$, $\begin{bmatrix} 0 \\ 3 \end{bmatrix}$, $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$.

Problem 11: Linear matrix expression

Let $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 2 & -2 \end{bmatrix}$. Find $3A + 2B$.

Solution 11

$$3A + 2B = \begin{bmatrix} 6 & 11 \\ 1 & 5 \end{bmatrix}.$$

Problem 12: Column picture

Let $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}$, $\mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$. Compute $A\mathbf{x}$ as a combination of columns.

Solution 12

$$A\mathbf{x} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 4 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 13 \\ -5 \end{bmatrix}.$$

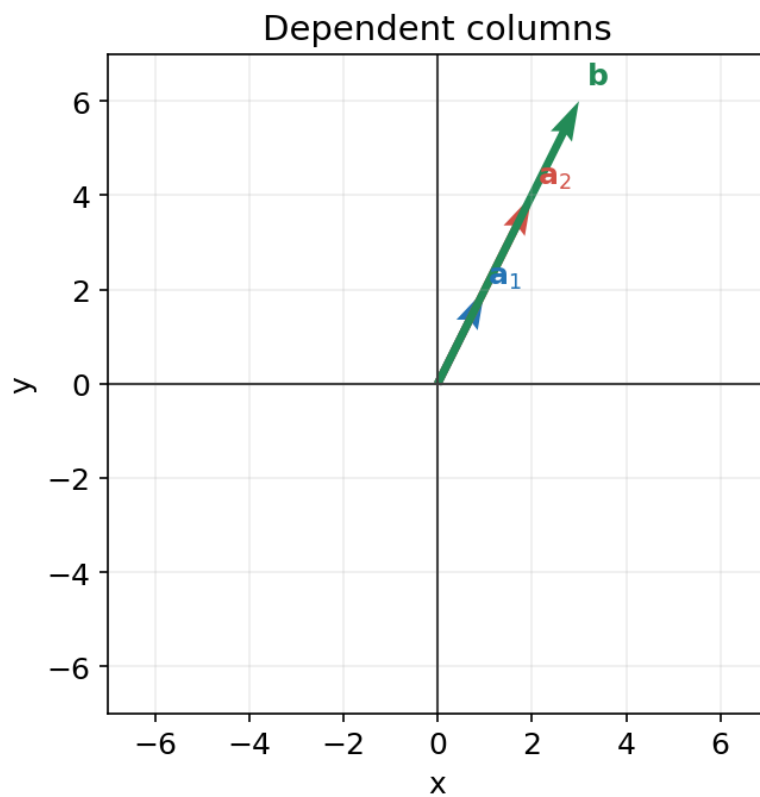
Problem 13: Find all weights

Let $\mathbf{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix}$. Find all $\mathbf{x}$ with $[\mathbf{a}_1 \ \mathbf{a}_2]\mathbf{x} = \mathbf{b}$.

Solution 13

The equation is $x_1 + 2x_2 = 3$. Thus

$$\mathbf{x} = \begin{bmatrix} 3 - 2t \\ t \end{bmatrix}, \quad t \in \mathbb{R}.$$



Problem 14: Challenge: cross-product identity

Without expanding coordinates, explain why $(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = 0$.

Solution 14

The cross product is orthogonal to both original vectors. Therefore

$$(\mathbf{u} \times \mathbf{v}) \cdot (\mathbf{u} + \mathbf{v}) = 0 + 0 = 0.$$

Problem 15: Challenge: recover a matrix

A is 2×2 . You know

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Find A and $A \begin{bmatrix} 5 \\ -2 \end{bmatrix}$.

Solution 15

The two images are the columns:

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}, \quad A \begin{bmatrix} 5 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ -13 \end{bmatrix}.$$